



Transitioning UPDM to the UAF

Matthew Hause (PTC)

Aurelijus Morkevicius Ph.D. (No Magic)

Graham Bleakley Ph.D. (IBM)

Co-Chairs OMG UPDM Group

OMG UAF Information day

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Agenda

- **Why do we need UPDM**
- **What was**
- **What is, and**
- **What will be**
- **Questions?**

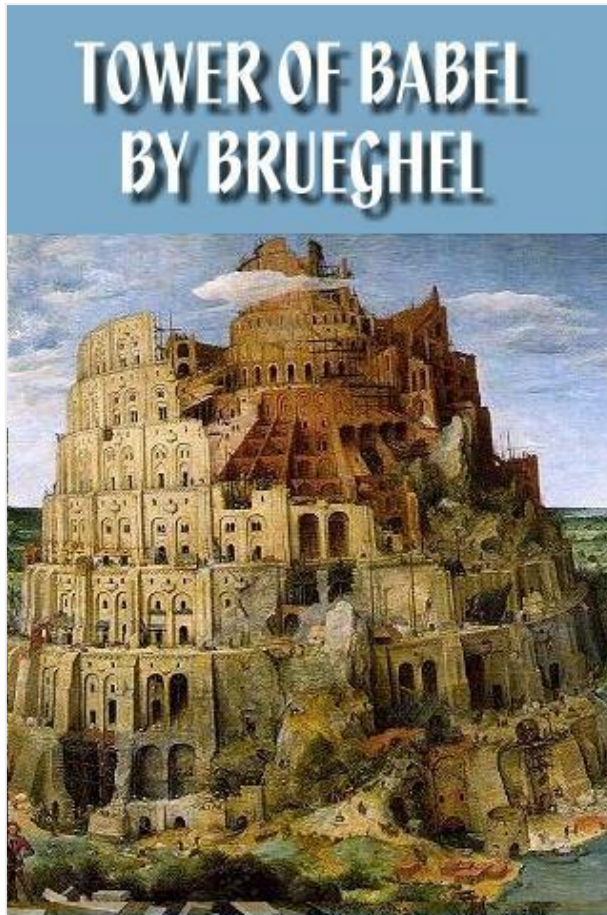


Why?

The Tower of Babel

A Communications Fable for our Time

Ancient



Modern



Does this solve the problem?

USA/UK: Two Countries Separated by a Common Language

- Even speaking the same language doesn't always help.

Picture this:

- A man wearing a vest, pants, and a pair of suspenders.

The American Image



The British Image



Vest

UK: Waistcoat

Suspenders

UK: Braces

Pants

UK: Trousers



So, if communication is hard with spoken language, are models the answer?

The Afghanistan Mission Network

Reference Document 3195

NATO Consultation, Command and Control Agency

Agence de Consultation, de Commandement et de Conduite des Opérations de l'OTAN



DEVELOPMENT OF THE AMN ARCHITECTURE IN 2010 – LESSONS LEARNED

Torsten Graeber, NATO C3 Agency

June 2011

The Hague



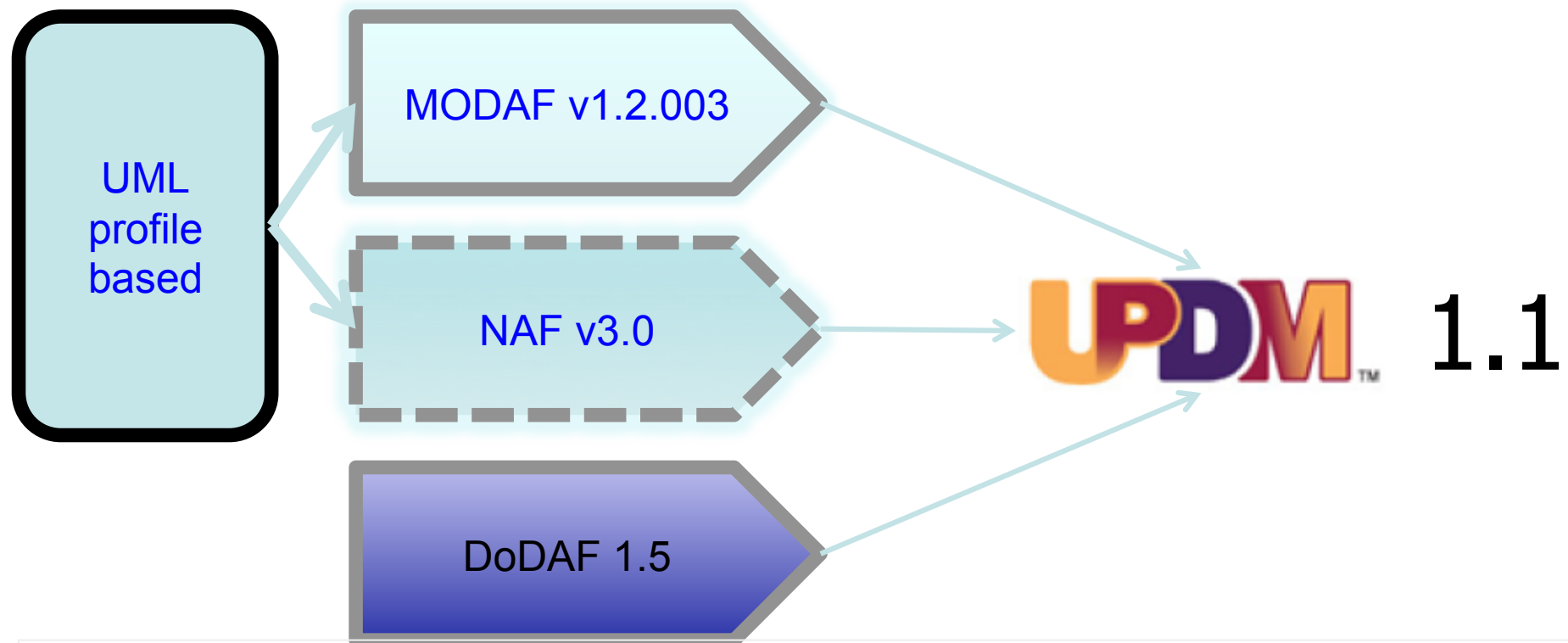
AMN Issues

- **These issues included:**
 - Different expectations on content and usage of the architecture leading to ever changing requirements and deliverables
 - No enforcement of the architecture during implementation
 - Usage of different architecture frameworks
 - Usage of different architecture tools.
 - No interchange between the tools
- **In late 2010, a governance structure for the AMN was endorsed by Chief Of Staff SHAPE and the AWG was included in this governance structure. As a direct consequence, the situation regarding clearer expectations, deliverables and enforcement of architecture has been improved in 2011.**
- **However, as the architects are sponsored by their respective nations they have to implement national policies and requirements, so that improvements regarding the usage of a single framework and tool are not to be expected.**



What was

UPDM version 1



- Meta model coherence
 - Same meta-model,
 - Different presentation layers
- Took an MBSE approach
- UPDM could choose between a pure UML or UML and SysML approach.
- UPDM contained both a profile and a domain meta-model

Why Model Based Systems Engineering

- **Pictures paints a thousand words**
 - Visio is good at this
 - Language is not controlled
- **Modeling languages add semantics and constraints**
 - Control what is being said and how it is said
- **MBSE is a common language of expression that captures**
 - Structure
 - Behaviour
 - Requirements
 - Functional
 - Non Functional
- **Models can be quantifiable and executable**



What is

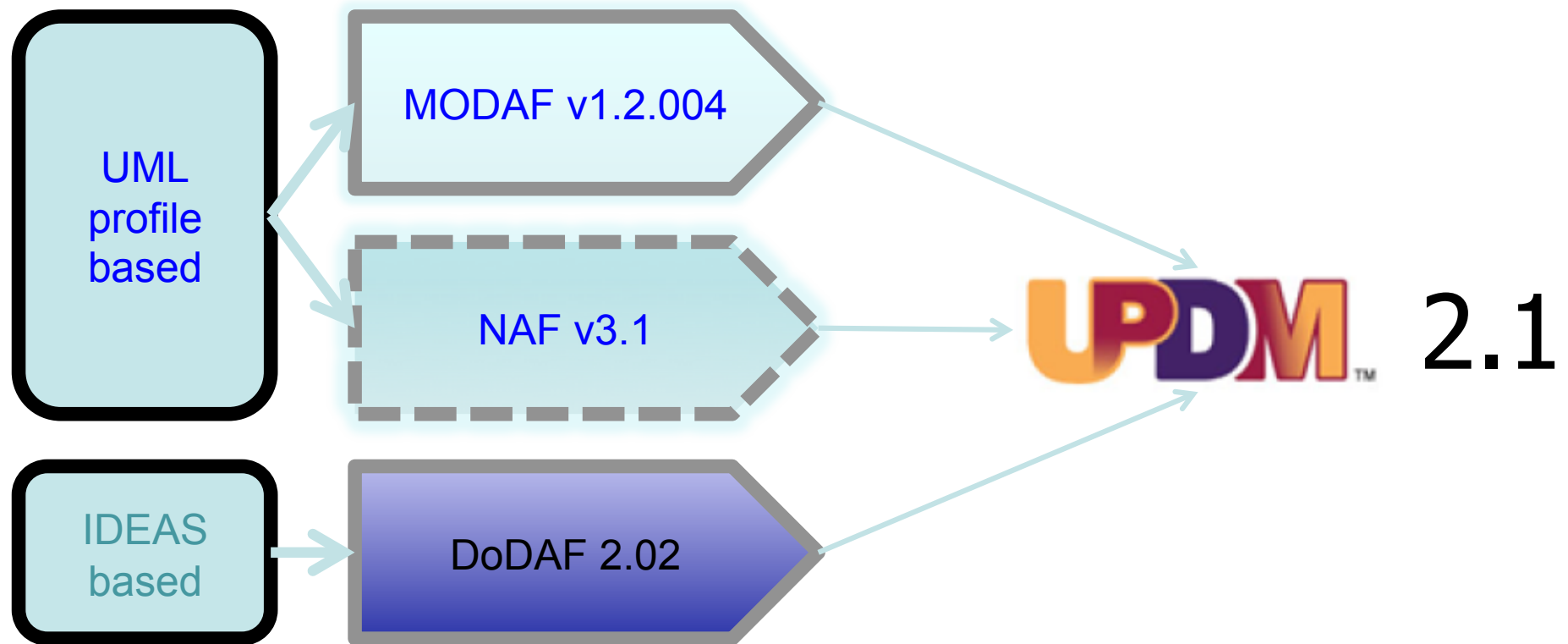
Current UPDM V 2.1

- UPDM is the **Unified Profile for DoDAF and MODAF + NAF (starting v2)**
- UPDM is **NOT** a new Architectural Framework
- UPDM is **NOT** a methodology or a process
- UPDM is a graphical **enterprise modeling language**
- UPDM was developed by members of the OMG with help from industry and government domain experts

- DOD (US)
- MOD (UK)
- SWAF (Swedish Armed Forces)
- DND (Canada)

- MITRE
- Raytheon
- Lockheed Martin
- General Dynamics
- L3

UPDM version 2 (2012-present day)



- IDEAS is a formal way for defining a metamodel
 - Allows you to reason across the information

IDEAS – International Defence Enterprise Architecture Specification

Supported by US, UK, SW, Australia, Canada

Unification with UPDM 2

- Common metamodel to build DoDAF, MODAF, and NAF models
 - Viewpoints (e.g.
Capability (DoDAF & NAF) vs.
Strategic (MODAF))
 - Views (e.g.
OV-2 Operational Resource Flow Description (DoDAF) vs.
OV-2 Operational Node Relationship Description (MODAF) vs.
NOV-2 Operational Node Connectivity Description (NAF))
 - Concepts (e.g.
Performer (DoDAF) vs.
Node (MODAF & NAF))
- Infrastructure for tools to be able to provide different environments for DoDAF, MODAF, NAF – underlying metamodel is the same
 - Common Meta-model, different presentation layers
- Easy transition among DoDAF, MODAF, and NAF models

MBSE and Engineering Analysis

Why UPDM is popular with practitioners of MBSE?

- No standardized frameworks for MBSE
- Integration with existing OMG standards, e.g. SysML, UML
 - **Common repository (Integrated Architecture Repository)**
 - **Application of engineering analysis methods**
 - Impact Analysis
 - Coverage Analysis
 - Trade-off Analysis
 - Behavioral execution
 - Requirements compliance analysis
 - Model-based testing
 - **Interoperability**

Adoption

- Tool Vendors: UPDM was adopted by majority of UML, SysML tool vendors.
- Defense:
 - Used by DOD and its contractors on various MBSE and IT projects
 - Being picked up outside of the US
 - Used in Europe, Australia, Asia, S. America
- Industry (external to Defense):
 - European research projects (DANSE)
 - Starting to be looked at by European industrial companies familiar with MBSE
 - Industry needs:
 - Commercialised/Industrialised whilst keeping features used by current users
 - Wider scope (SoS Lifecycle, Human System Integration, Risk etc.)



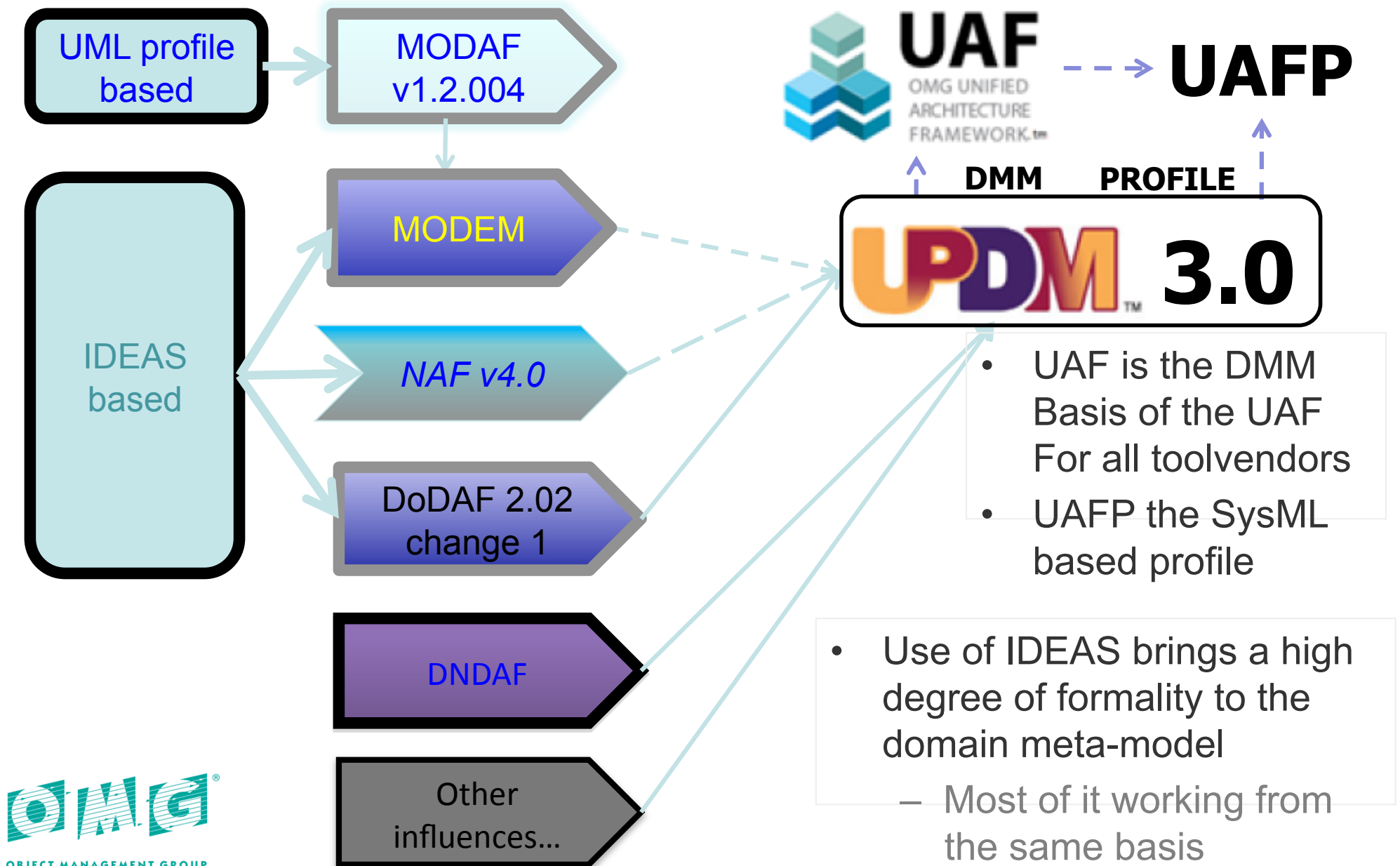
What will be UPDM 3-> UAF 1.0

Framework developments

- UPDM RFP requirement: " The UPDM V3.0 domain metamodel shall be derived from MODEM and DM2, both of which are based upon the International Defence Enterprise Architecture Specification Foundation [IDEAS]."
 - **Mandatory requirements (excerpt):**
 - Provide Domain Metamodel derived from MODEM and DM2 ✓
 - An Architecture Framework Profile Using SysML ✓
 - Supports BPMN 2.0 ✓
 - Use of SysML Requirements Elements and Diagrams ✓
 - Use of SysML Parametric Elements and Diagrams Mapped to Measurements ✓
 - Traceability Matrix to Supported Frameworks ✓
 - **Non mandatory features (excerpt):**
 - UML Profile for NIEM ✓
 - Information Exchange Packaging Policy Vocabulary (IEPPV) ✓
 - Viewpoints in Support of SoS Life Cycle Processes and Analyses ✓
 - Support for Fit for Purpose Viewpoints beyond those defined in DoDAF, MODAF/ MODEM, NAF, and the Security Viewpoint from DNDAF. ✓
 - Human Systems Integration (HSI) ✓



UPDM version 3



Why a Unified Architecture Framework

- Proliferation of frameworks that UPDM was being asked to support
- Need to support industry and federal usage as well as military
 - Commercialisation, whilst still supporting Warfighter needs
- Ability to support other frameworks
 - By Extension
 - By Mapping
- IDEAS based format for DMM Allows implementation by non-SysML based tools
 - Same format as DoDAF 2.0.2 Change 1

Grid Approach

	Behaviour								
	Classification	Structure	Connectivity	Processes	States	Sequences	Information	Constraints	Programme
Enterprise	E1 Capability Taxonomy <i>NAV-2, NCV-2 AV-2, StV-2</i>	E2 Enterprise Vision <i>NCV-1 StV-1</i>	E3 Capability Dependencies <i>NCV-4 StV-4</i>	E4 Standard Processes <i>NCV-6 StV-6</i>	E5 Effects		E7 Performance Parameters <i>NCV-1 StV-1</i>	E8 Planning Assumptions	Ep Capability Phasing <i>NCV-3 StV-3</i>
Service	E1-S1 (NSOV-3) S1 Service Taxonomy <i>NAV-2, NSOV-1 AV-2, SOV-1</i>		S3 Service Interfaces <i>NSOV-2 SOV-2</i>	S4 Service Functions <i>NSOV-3 SOV-5</i>	S5 Service States <i>NSOV-4b SOV-4b</i>	S6 Service Interactions <i>NSOV-4c SOV-4c</i>	S7 Service I/F Parameters <i>NSOV-2 SOV-2</i>	S8 Service Policy <i>NSOV-4a SOV-4a</i>	Sp Service Delivery
Logical	L1 Node Types <i>NAV-2 AV-2</i>	L2 Logical Scenario <i>NOV-2 OV-2</i>	L3 Node Interactions <i>NOV-2, NOV-3 OV-2, OV-3</i>	L4 Logical Activities <i>NOV-5 OV-5</i>	L5 Logical States <i>NOV-6b OV-6b</i>	L6 Logical Sequence <i>NOV-6c OV-6c</i>	L7 Logical Data Model <i>NSV-11a OV-7</i>	L8 Logical Constraints <i>NOV-6a OV-6a</i>	Lp Lines of Development <i>NPV-2 AcV-2</i>
Resources	R1 Resource Types <i>NAV-2, NSV-9 AV-2, SV-9</i>	R2 Resource Structure <i>NOV-4, NSV-1 OV-4, SV-1</i>	R3 Resource Connectivity <i>NSV-2, NSV-6 SV-2, SV-6</i>	L4-R4 (NSV-5) R4 Resource Functions <i>NSV-4 SV-4</i>	R5 Resource States <i>NSV-10b SV-10b</i>	R6 Resource Sequence <i>NSV-10c SV-10c</i>	R7 Physical Data Model <i>NSV-11b SV-11</i>	R8 Resource Constraints <i>NSV-10a SV-10a</i>	Rp Configuration Management <i>NSV-8 SV-8</i>
Deployed	D1 Master Data <i>NAV-2 AV-2</i>	D2 Deployed Resources <i>NCV-5, NOV-4 StV-5, OV-4</i>							Dp Deployment Schedule <i>NCV-5 StV-5</i>
Architecture	A1 Meta-Data Definitions <i>NAV-3 AV-1/2</i>	A2 Architecture Products	A3 Architecture Correspondance <i>ISO42010</i>	A4 Methodology Used <i>NAF Ch3</i>	A5 Architecture Status <i>NAV-1 AV-1</i>	A6 Architecture Versions <i>NAV-1 AV-1</i>	A7 Architecture Meta-Data <i>NAV-1/3 AV-1</i>	A8 Standards <i>NTV-1/2 TV-1/2</i>	Ap Architecture Plan

Why the Grid ?

- Very hard to manage the views with so many contributing frameworks
 - Lead to very complex mapping tables
 - Unwieldy descriptions
- Provides an abstraction layer so it is possible to map many other frameworks onto the MM
 - HSI views and SoS Lifecycle views
- Commercialises the UAF whilst supporting Warfighter needs
 - Still the same underlying architectural data structures and view constructs that support
 - DoDAF
 - MODAF/MODEM
 - NAF
 - Same data model, different presentation layer

Conclusions

- UAF has the potential to improve communication, collaboration and interoperability between
 - Nations
 - Government and Industry
 - Industry to Industry
- Grid approach allows different industries to reuse, extend or create new views appropriate to them (Fit for purpose)
- New technologies can and will be applied to extend the use of UAF architectures to enable
 - Architecture Federation
 - Tool Federation
 - Improved interoperability
- Improving the discovery and reuse of architectural artifacts